

BC220418499: ARSLAN SAEED

MTH501 - Linear Algebra (Quiz 3)

Question # 8 of 10 (Start time: 10:22:43 PM, 27 June 2024)

Multiplication of a partitioned matrix by a scalar is also computed __

Select the correct option

○	Column by column
○	Diagonal by diagonal
○	Block by block
○	Row by row

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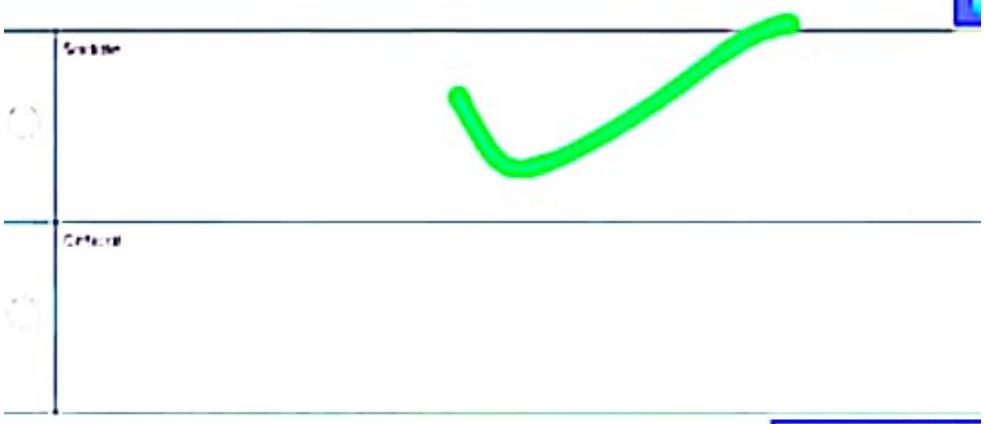
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MTH501 - Linear Algebra (Quiz 3) Quiz Start T

uestion # 6 of 10 (Start time: 10:20:39 PM, 27 June 2024)

If a matrix A has both negative and positive eigen values, so in this case origin behaves as a ---------- point

elect the correct option



Click to Save Answer

If A is invertible and b in R^n be any vector. Then we must have a unique $A^{-1}b$, which is a solution of .....

Select the correct option

○	$A^2x = b$
○	$A^{-1}x = b$
○	$A^2x - b$
○	$Ax = b$

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MTH501 - Linear Algebra (Quiz 3)

Question # 5 of 10 (Start time: 10:19:54 PM, 27 June 2024)

We can add the matrices of _____

Select the correct option

- different order
- same number of rows
- same number of columns
- same order

CamScanner

TH501 - Linear Algebra (Quiz 3) Quiz Start Time: 09:30 PM, 27 June 202

Question # 1 of 10 (Start time: 09:30:38 PM, 27 June 2024) Total Mark

If A be $n \times n$ matrix, then $det(A^T) =$ .....

Select the correct option

Reload Math Equations

- Inverse of A
- 1 / det A
- $det A^T$
- det A

Click to Save Answer & Move to Next Question

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Question # 10 of 10 (Start time: 09:22:56 PM, 27 June 2024) Total Marks:

The determinant of a triangular matrix is the sum of the entries of the main diagonal.

Select the correct option

Reload Math Equations

- False
- True

Click to Save Answer & Move to Next Question

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Which of the following is (are) the Eignvalue(s) of the matrix?

$$A = \begin{bmatrix} 3 & 0 \\ \backslash 5 & 3 \end{bmatrix}$$

elect the correct option

Reload Math Equations

- ○ 3
- ○ 5
- ○ 3, 4
- ○ 3, 5

Click to Save Answer & Move to Next Question

CamScanner

Question # 5 of 10 (Start time: 09:18:40 PM, 27 June 2024) Total Marks

What is Eigen value?

Select the correct option

- ○ It is the inverse of the transform
- ○ A scalar associated with a given linear transformation
- ○ A vector obtained from the coordinates
- ○ A matrix determined from the algebraic equations

Click to Save Answer & Move to Next Question

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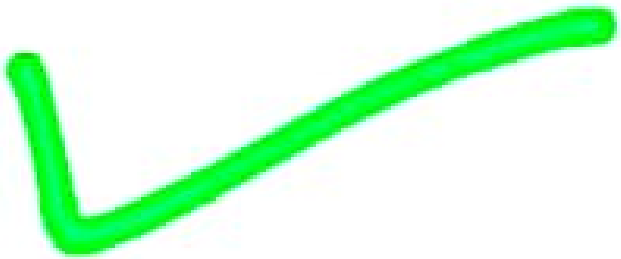
uestion # 4 of 10 (Start time: 09:34:54 PM, 27 June 2024) Total Mark

Two vectors are linearly dependent if and only if they lie_____

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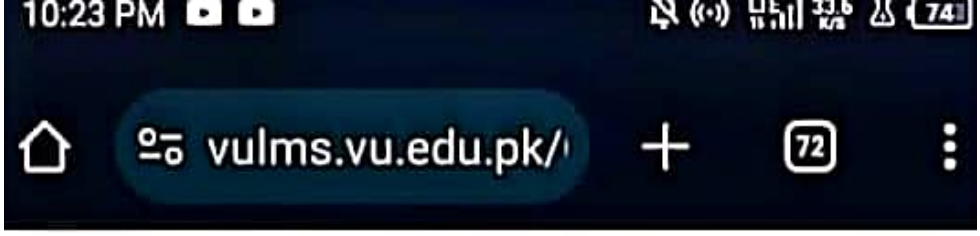
Reload Math Equations

- on a line parallel to x-axis
- on the same line through origin



Click to Save Answer & Move to Next Question

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MTH501 - Linear Algebra (Quiz 3)

Question # 8 of 10 (Start time: 10:22:43 PM, 27 June 2024)

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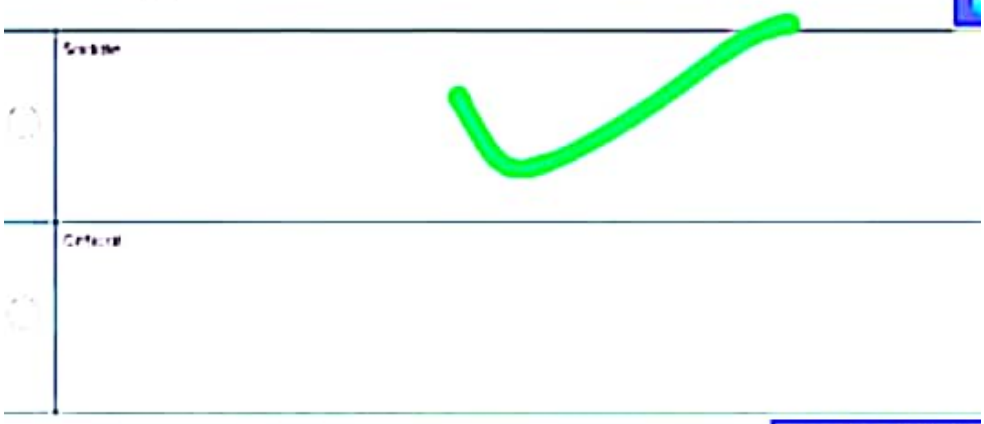
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We can add the matrices of _____

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- same number of rows
- same number of columns
- same order

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Question # 3 of 10 (Start time: 09:34:26 PM, 27 June 2024) Total Mark:

If A^T transpose of A(where A is a square matrix) then which of the following is the most appropriate option for A?

Select the correct option

Reload Math Equations

- A is a scalar matrix.
- A is symmetric matrix.
- A is singular matrix.
- A is invertible matrix.

Click to Save Answer & Move to Next Question

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Question # 8 of 10 (Start time: 09:38:13 PM, 27 June 2024)
Total Marks

Eigenvalues of a square matrix are always

Select the correct option

- Positive
- Real and Imaginary
- Negative
- Real

Question # 7 of 10 (Start time: 09:36:46 PM, 27 June 2024) Total Marks

If $\lambda - 2$ is a factor of the characteristic polynomial of matrix A, then which of the following is the Eigenvalue of A ?

Select the correct option

Reload Math Equations

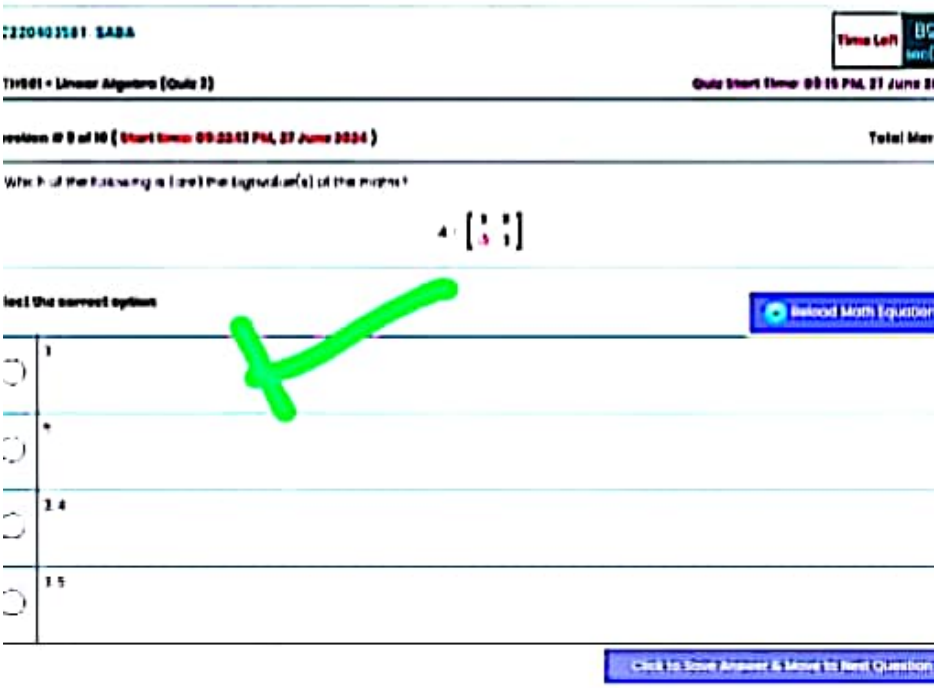
- 2
- $\frac{1}{2}$
- 0
- -2

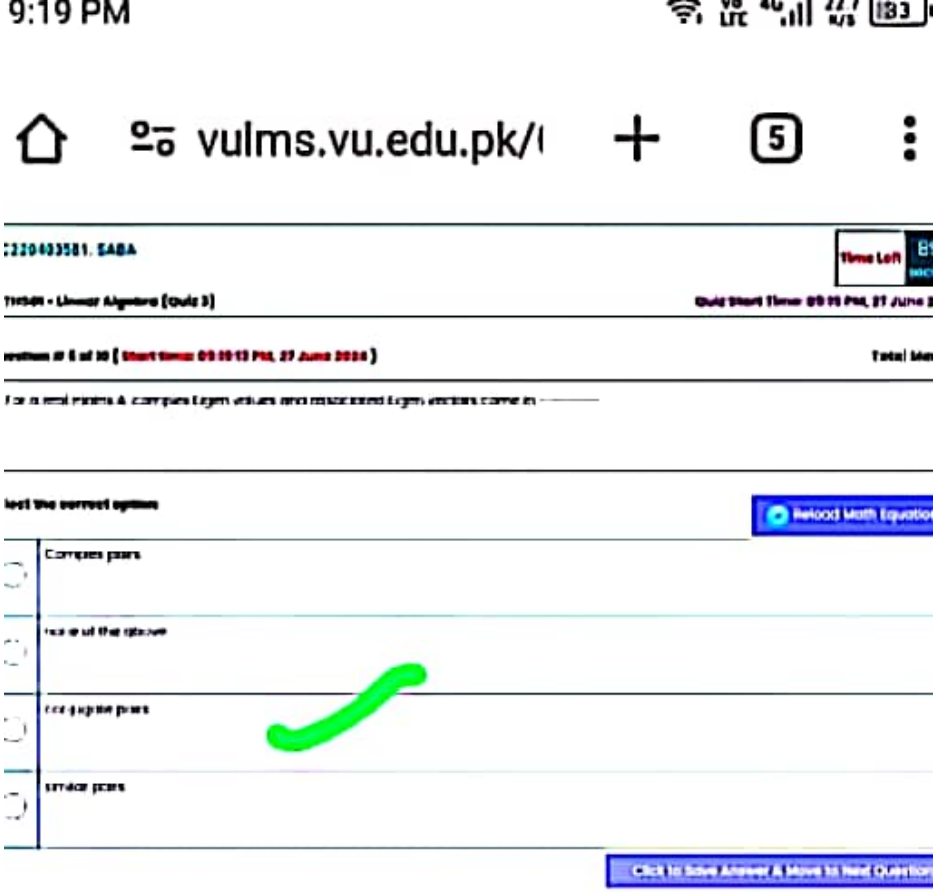
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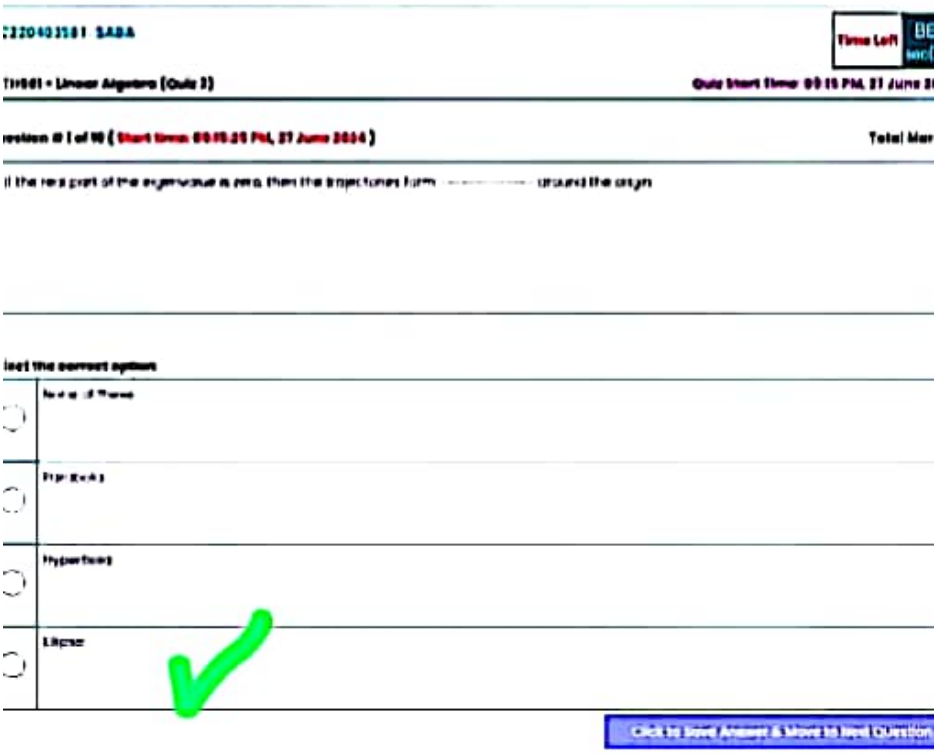
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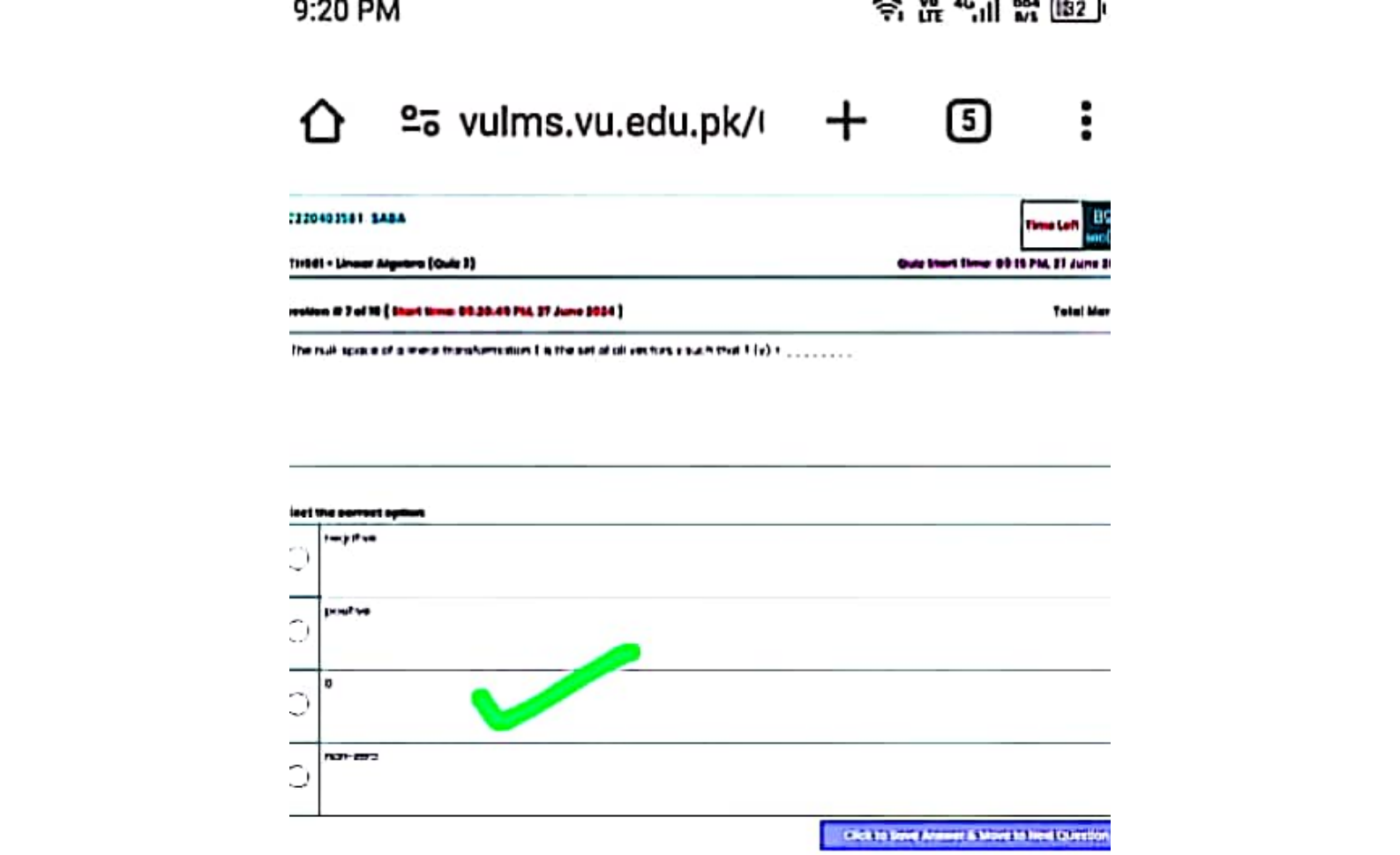

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Reload Math Equation
Complex pairs
similar pairs
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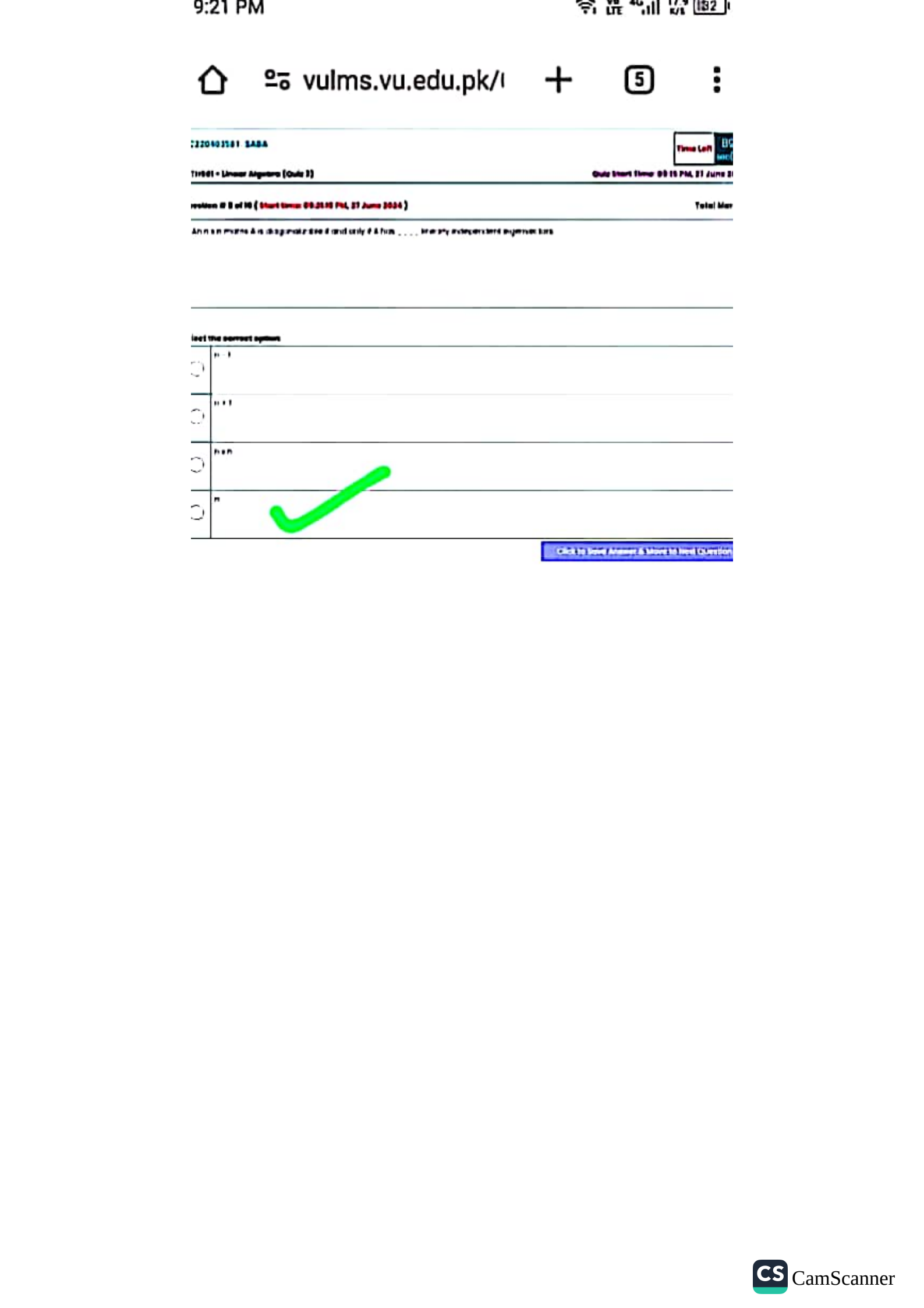
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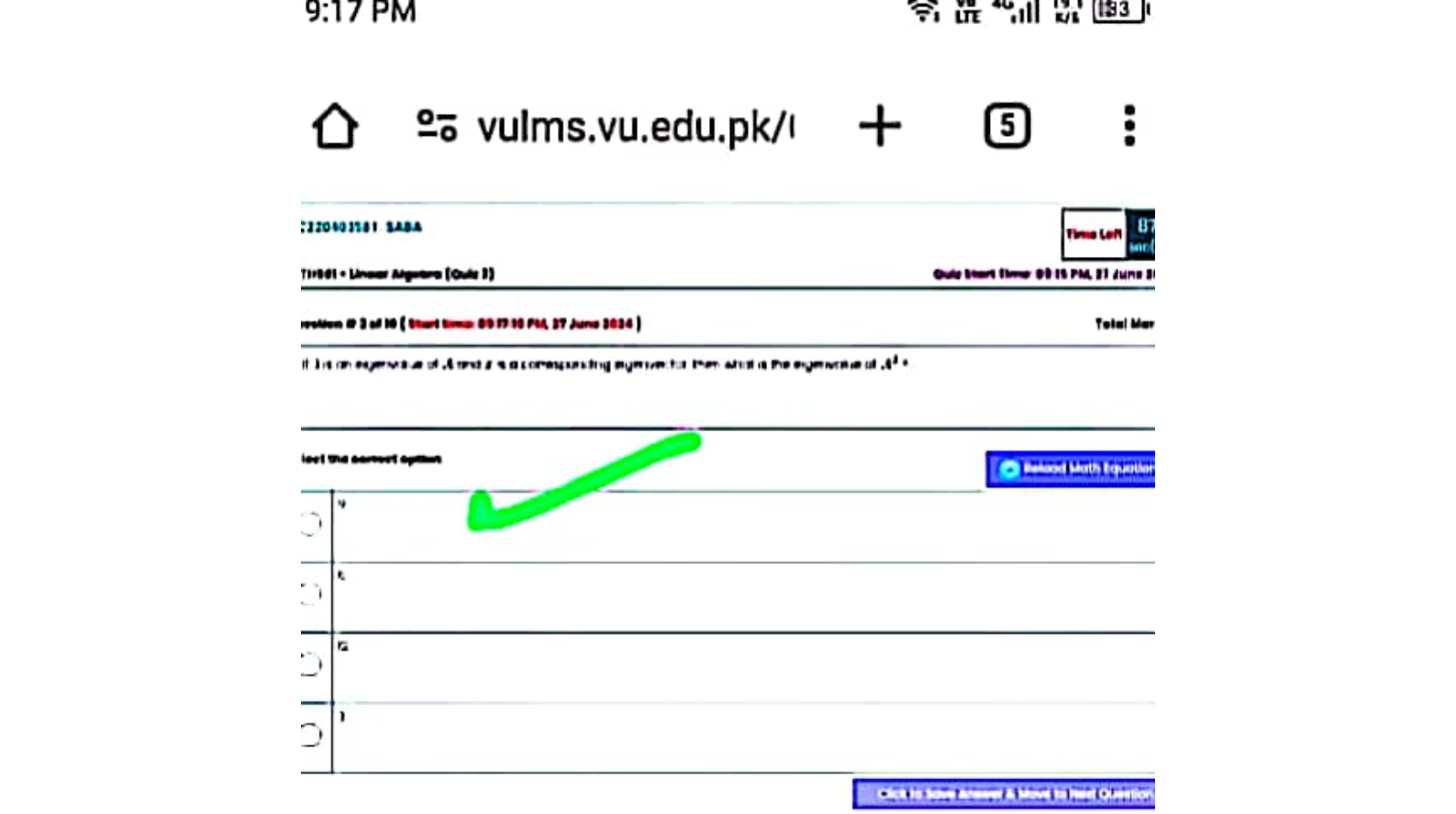


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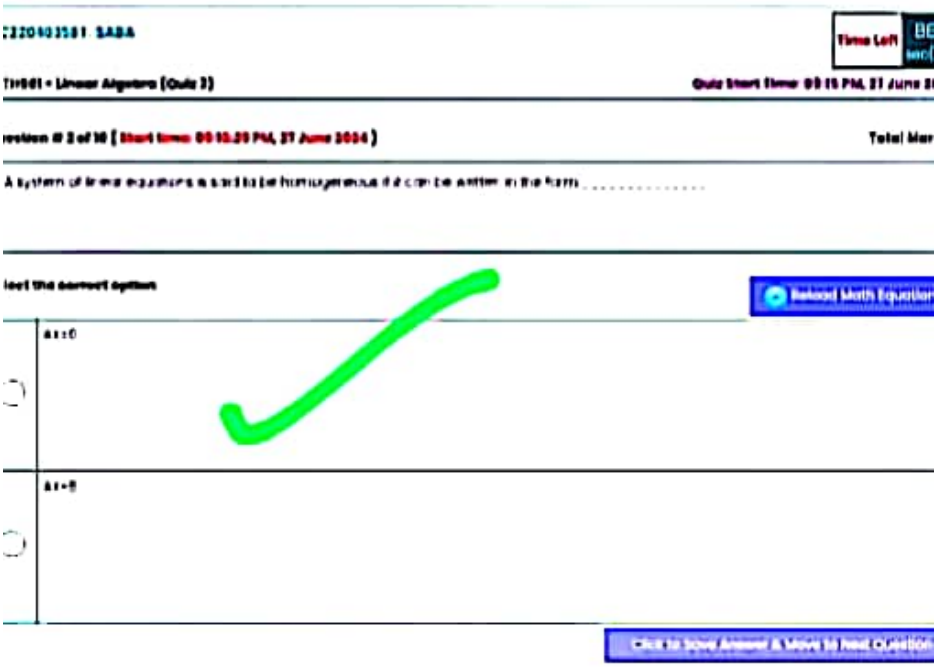
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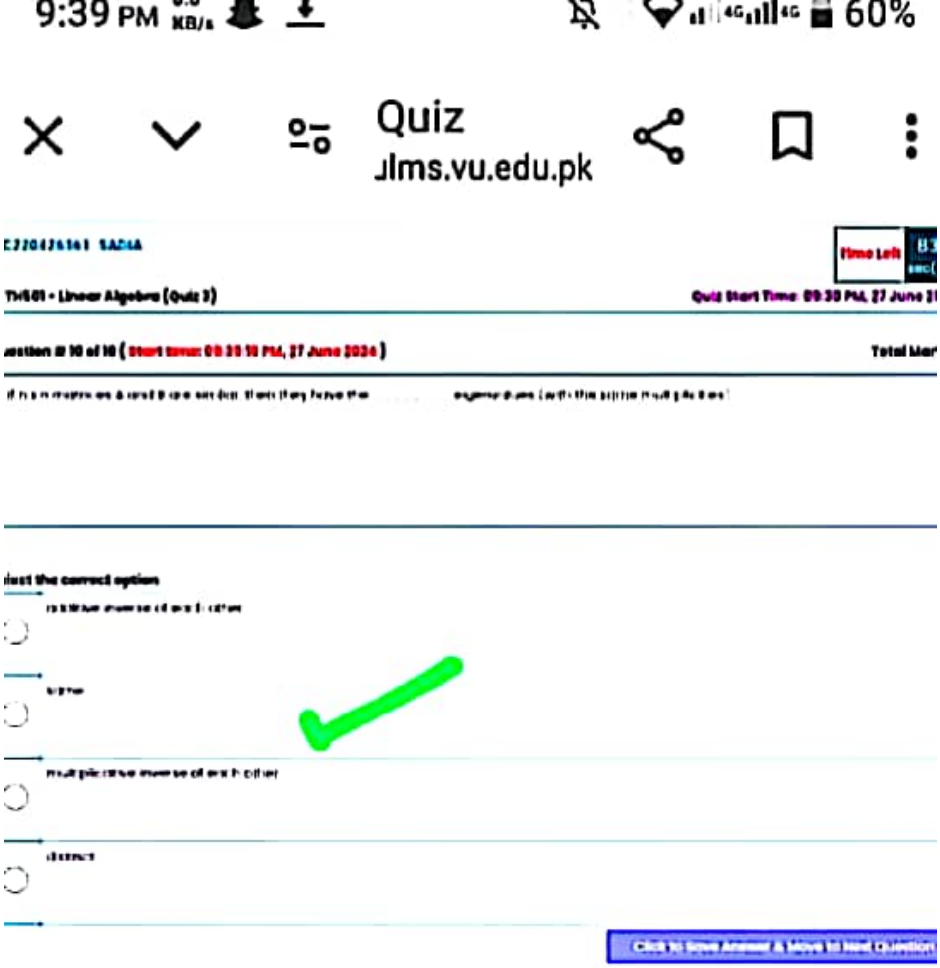

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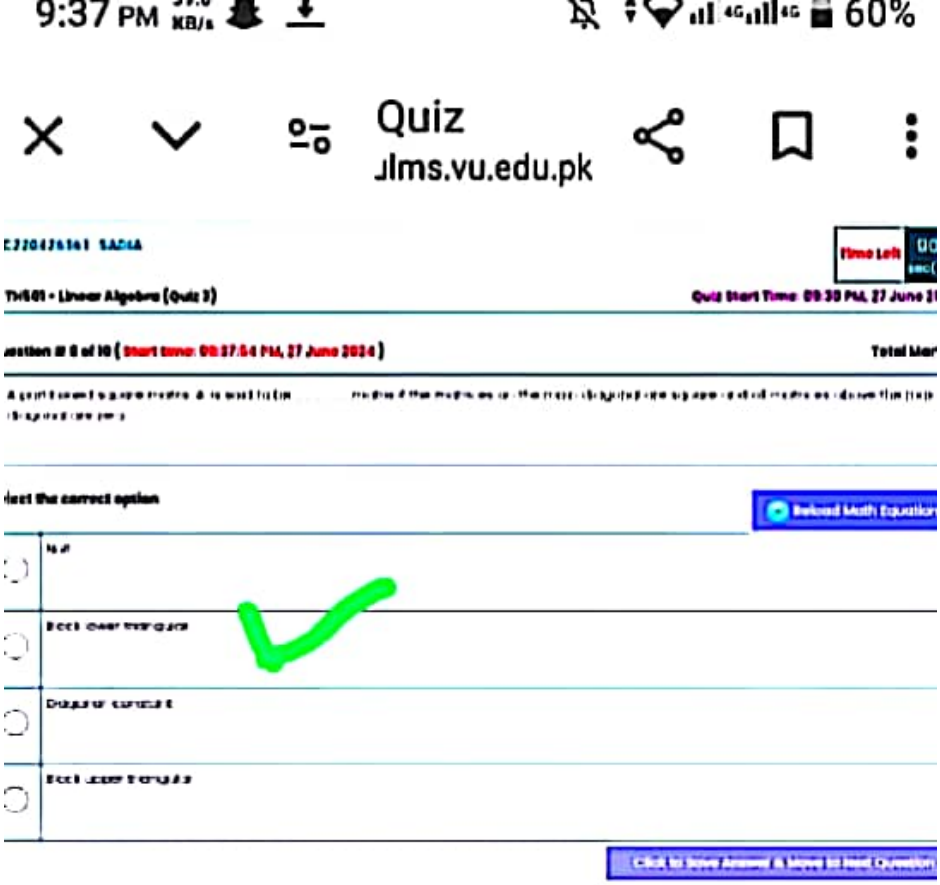
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Quiz Start Time:
Total Mar
Click to Save Answer & Move to Next Question

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Question # 5 of 10 (Start time: 09:35:21 PM, 27 June 2024) Total Marks

If one of the eigenvalues of $[A]_{n\times n}$ is zero it implies --------

Select the correct option

Reload Math Equations

- The solution to $[A][X] = [C]$ a system of equations is unique
- The determinant of $[A]$ is nonzero
- The determinant of $[A]$ is zero
- The solution to $[A][X] = [0]$ system of equations is trivial

Click to Save Answer & Move to Next Question

Not confirm

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Question # 7 of 10 (Start time: 09:36:46 PM, 27 June 2024) Total Marks

If $\lambda - 2$ is a factor of the characteristic polynomial of matrix A, then which of the following is the Eigenvalue of A ?

Select the correct option

Reload Math Equations

- 2
- $\frac{1}{2}$
- 0
- -2

Click to Save Answer & Move to Next Question

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Question # 8 of 10 (Start time: 09:38:13 PM, 27 June 2024)
Total Marks

Eigenvalues of a square matrix are always

Select the correct option

- Positive
- Real and Imaginary
- Negative
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Question # 3 of 10 (Start time: 09:34:26 PM, 27 June 2024) Total Mark:

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- A is singular matrix.
- A is invertible matrix.

Click to Save Answer & Move to Next Question

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TH501 - Linear Algebra (Quiz 3) Quiz Start Time: 09:30 PM, 27 June 202

Question # 1 of 10 (Start time: 09:30:38 PM, 27 June 2024) Total Mark

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Select the correct option

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- 1 / det A
- $det A^T$
- det A

Click to Save Answer & Move to Next Question

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Question # 10 of 10 (Start time: 09:22:56 PM, 27 June 2024) Total Marks:

The determinant of a triangular matrix is the sum of the entries of the main diagonal.

Select the correct option

Reload Math Equations

- False
- True

Click to Save Answer & Move to Next Question

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Which of the following is (are) the Eignvalue(s) of the matrix?

$$A = \begin{bmatrix} 3 & 0 \\ \backslash 5 & 3 \end{bmatrix}$$

elect the correct option

Reload Math Equations

- ○ 3
- ○ 5
- ○ 3, 4
- ○ 3, 5

Click to Save Answer & Move to Next Question

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Question # 5 of 10 (Start time: 09:18:40 PM, 27 June 2024) Total Marks

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- ○ A vector obtained from the coordinates
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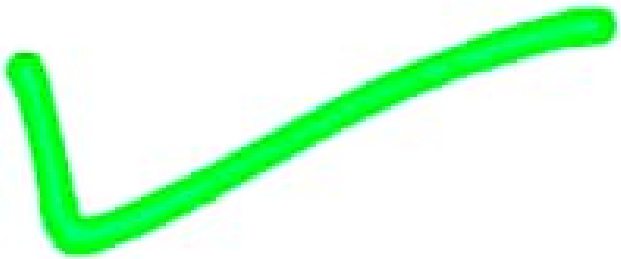
uestion # 4 of 10 (Start time: 09:34:54 PM, 27 June 2024) Total Mark

Two vectors are linearly dependent if and only if they lie_____

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Reload Math Equations

- on a line parallel to x-axis
- on the same line through origin



Click to Save Answer & Move to Next Question

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MTH501 – Linear Algebra (Quiz 3)

Question # 5 of 10 (Start time: 11:07:27 PM, 27 June 2024)

If n x n matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

Select the correct option

- False
- True ✓

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ITH501 - Linear Algebra (Quiz 3)

Question # 8 of 10 (Start time: 11:08:53 PM, 27 June 2024)

When a matrix A has a complex eigenvalue with positive real part, the trajectories spiral ----------

Select the correct option

- (●) Outwards
- ( ) Inwards

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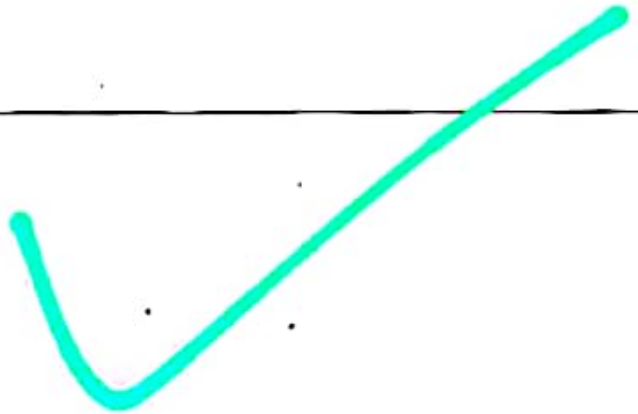
MTH501 – Linear Algebra (Quiz 3)

Question # 1 of 10 (Start time: 11:03:21 PM, 27 June 2024)

Electric circuit rotation is caused by sine and cosine functions if there eigenvalues are complex and hence the origin is called ------

Select the correct option

○	Critical point
○	Spiral point



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Question # 6 of 10 (Start time: 11:08:06 PM, 27 June 2024)

A system of linear equations is said to be homogeneous if it can be written in the form _____

Select the correct option

- (●) AX=0
- ( ) AX=B

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Question # 3 of 10 (Start time: 11:05:25 PM, 27 June 2024)

$$x_1 + 2x_2 + 3x_3 = 7$$

$$4x_1 + x_2 + 2x_3 = 2$$

$$-4x_1 + 3x_2 + 9x_3 = 4$$

The augmented matrix for the system is ----------

Select the correct option

$$\begin{bmatrix} 1 & 2 & 3 & 7 \\ -4 & 1 & 2 & 2 \\ 4 & 3 & 9 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 & 7 \\ 4 & 1 & 2 & 2 \end{bmatrix}$$

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Question # 10 of 10 (Start time: 11:11:31 PM, 27 June 2024)

The complex conjugate of a complex vector x in C^n is the vector $\bar{x}$ in C^n whose entries are the ---------- conjugates of the entries in x.

Select the correct option

- (●) Complex
- ( ) Real

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MTH501 - Linear Algebra (Quiz 3)

Question # 4 of 10 (Start time: 11:06:11 PM, 27 June 2024)

Let A be n x n matrix, then A is invertible if and only if

Select the correct option

- ○ det A is not zero
- ○ det A is zero

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MTH501 - Linear Algebra (Quiz 3)

Question # 9 of 10 (Start time: 11:10:13 PM, 27 June 2024)

ϕ is the angle between the positive x-axis and the ray

Select the correct option

○	From (a,b) through (0,0)
○	From (1,1) through (0,0)
○	None of these
○	From (0,0) through (a,b)

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Question # 7 of 10 (Start time: 11:08:32 PM, 27 June 2024)

If x - 2 is a factor of the characteristic polynomial of matrix C then an eigenvalue of C is

Select the correct option

○	0
○	-2
○	1/2
◉	2

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Question # 2 of 10 (Start time: 11:04:00 PM, 27 June 2024)

The eigenvalues of a 4 by 4 matrix [A] are given as 2, -3, 1, 3 and 7. The $|det(A)|$ then is

Select the correct option

- ○ 530
- ○ 546
- ○ cannot be determined
- ○ 25

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MTH501 - Linear Algebra (Quiz 3)

Question # 1 of 10 (Start time: 05:08:33 AM, 27 June 2024)

What is Eigen value?

Select the correct option

- ○ It is the inverse of the transform
- ◉ A scalar associated with a given linear transformation
- ○ A vector obtained from the coordinates
- ○ A matrix determined from the algebraic equations

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MTH501 - Linear Algebra (Quiz 3)

Question # 3 of 10 (Start time: 05:10:22 AM, 27 June 2024)

Let $T: R^2 \to R^2$ be transformation $T(x_1, x_2) = (x, 0)$. The null space (or kernel) $N(T)$ of T is

Select the correct option

- $(x_1, 0)$
- $(0, x_2)$*
- $(1, 0)$
- $(0, 1)$

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MTH501 - Linear Algebra (Quiz 3)

Question # 4 of 10 (Start time: 05:11:12 AM, 27 June 2024)

The null-space of a linear transformation T is the set of all vectors v such that T (v) = _____.

Select the correct option

○	non-zero
●	0
○	positive
○	negative

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MTH501 - Linear Algebra (Quiz 3)

Question # 5 of 10 (Start time: 05:11:49 AM, 27 June 2024)

What is the maximum possible number of pivots in a 6x6 matrix?

Select the correct option

- 2
- 6
- 4
- 0

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MTH501 - Linear Algebra (Quiz 3)

Question # 7 of 10 (Start time: 05:12:55 AM, 27 June 2024)

Electric circuit rotation is caused by sine and cosine functions if there eigenvalues are complex and hence the origin is called --------

Select the correct option

- (●) Spiral point
- ( ) Critical point

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MTH501 - Linear Algebra (Quiz 3)

Question # 8 of 10 (Start time: 05:14:10 AM, 27 June 2024)

Two equivalent vectors must have the same initial point:

Select the correct option

- ○ May be
- ○ true
- ○ None of the above
- ● false

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MTH501 - Linear Algebra (Quiz 3)

Question # 10 of 10 (Start time: 05:15:34 AM, 27 June 2024)

If λ is an eigenvector of A, then every nonzero vector x such that $Ax = \lambda x$ is called an ---------- of A corresponding to ----

Select the correct option

- Eigenvalue, A
- Eigenvector, λ
- Eigenvector, A
- Eigenvalue, λ

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MTH501 - Linear Algebra (Quiz 3)

Question # 9 of 10 (Start time: 05:14:59 AM, 27 June 2024)

If n x n matrices A and B are similar, then they have the _____ characteristic polynomial.

Select the correct option

- ○ different
- ○ inverse of the other
- ◉ same
- ○ same but opposite in sign

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Question # 6 of 10 (Start time: 05:12:26 AM, 27 June 2024)

An $n \times n$ matrix A is said to be diagonalizable if and only if A has n ---------- eigenvectors.

Select the correct option

○	Linearly dependent
◉	Linearly Independent

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MTH501 - Linear Algebra (Quiz 3)

Question # 2 of 10 (Start time: 05:09:20 AM, 27 June 2024)

The complex conjugate of a complex vector x in C^n is the vector $\bar{x}$ in C^n whose entries are the ---------- conjugates of the entries in x.

Select the correct option

- ○ Real
- ● Complex

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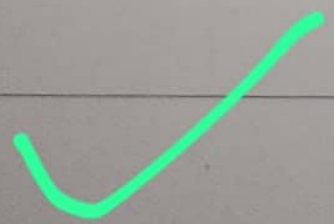
MTH501 - Linear Algebra (Quiz 3)

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- ○ 1/2
- ◉ 2



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MTH501 - Linear Algebra (Quiz 3)

Question # 9 of 10 (Start time: 11:10:13 PM, 27 June 2024)

ϕ is the angle between the positive x-axis and the ray

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- ○ From (a,b) through (0,0)
- ○ From (1,1) through (0,0)
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MTH501 - Linear Algebra (Quiz 3)

Question # 4 of 10 (Start time: 11:06:11 PM, 27 June 2024)

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Question # 6 of 10 (Start time: 11:08:06 PM, 27 June 2024)

A system of linear equations is said to be homogeneous if it can be written in the form _____

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Select the correct option

- False
- True ✓

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